

Terrain-Adaptive Reinforcement Learning for Hexapod Locomotion

Naga Sri Chaitnaya Balusu
Lithin Pakide
Seth Jones
Benjamin Mayfield
Mohammed Aijaz Ahmed Khan
Ronin Nelson
Christian Mutengela

¹School of Informatics, Computing and Cyber Systems, Northern Arizona University

²School of Informatics, Computing and Cyber Systems, Northern Arizona University

Abstract

Legged robots exhibit significant performance degradation when transitioning across surfaces with different stiffness and damping properties, largely due to changes in contact mechanics and ground reaction forces. This work investigates a reinforcement learning (RL) framework for adaptive locomotion on a low-cost hexapod robot without external computation or simulation. A discrete action space comprising step height, stride length, and gait speed was evaluated using a tabular Q-learning algorithm. Surface classification was derived from pitch-roll variance using an onboard IMU, enabling selection of terrain-appropriate gait parameters. Experiments conducted on hard floor, carpet, and cushion surfaces demonstrate that the learned gait policy yields measurable improvements in forward displacement and stability relative to a fixed baseline gait. The resulting Q-value distribution reveals structured clusters, with optimal gaits consistently associated with higher step height and moderate stride length. The study shows that lightweight RL techniques can enable terrain-adaptive locomotion on embedded hardware platforms and provides a reproducible methodology for future studies in hexapod control.

1 Introduction

Hexapod locomotion has long been recognized as a robust approach for navigating unstructured terrain due to its inherent static stability, redundancy, and fault tolerance (1). However, even with stable tripod or wave gaits, performance typically degrades when moving across surfaces with variable compliance such as carpet, foam, or deformable soil. These surfaces introduce increased damping, loss of traction, and reduced effective stride length, leading to inefficient locomotion, excessive oscillation, and in severe cases locomotion failure.

Traditional approaches to terrain-adaptive walking rely on either: (i) detailed mechanical modeling of leg-terrain interactions (2), (ii) heuristic rules derived from empirical tuning (3), or (iii) high-dimensional optimization in simulation (4). Model-based approaches require accurate physical parameters that are difficult to obtain for soft or deforming surfaces. Heuristic approaches are highly platform-specific and fail to generalize. Simulation-trained deep reinforcement learning (DRL) has become prominent in quadruped locomotion due to its ability to learn complex behaviors (5; 6). However, DRL typically requires high computational resources, realistic simulation-to-reality transfer, dense sensor feedback, and continuous action spaces.

These assumptions do not hold for educational hexapod platforms driven by a Raspberry Pi controller and low-cost servos. In contrast, discrete RL methods such as tabular Q-learning offer computational simplicity while retaining the ability to learn surface-specific gait adaptations (7). When the gait space is parameterized appropriately, Q-learning can efficiently identify robust locomotion strategies without requiring simulation or external sensors.

The key research question addressed in this work is:

Can a lightweight reinforcement learning controller discover gait configurations that maintain walking performance across surfaces of varying stiffness on a low-cost hexapod robot?

To answer this question, a reproducible RL framework is developed that:

- discretizes gait parameters into 27 actions,
- classifies terrain using IMU variance,
- computes reward from forward displacement and stability,
- updates a 4×27 Q-table in real time,

- operates entirely on embedded hardware.

A photograph of the experimental platform is shown in Fig. 1.



Figure 1: The Freenove Big Hexapod Robot used in this study. The robot features 18 DOF, onboard IMU, and Raspberry Pi control.

2 Methodology

2.1 Hardware Platform

The robot used in this study is the Freenove Big Hexapod Robot Kit, equipped with:

- 18 MG996R servos (stall torque ≈ 10 kg cm),

- PCA9685 PWM driver (50–330 Hz),
- MPU-6050 IMU providing 3-axis accelerometer and gyroscope data,
- HC-SR04 ultrasonic sensor for obstacle detection,
- Raspberry Pi 4 running Python with a 50–60 Hz control loop.

Servo calibration offsets were determined experimentally and stored in `point.txt`. These offsets compensate for mechanical asymmetries and ensure consistent neutral poses.

2.2 Gait Parameterization

Each gait action is defined by a tuple (Z, Y, S) :

$$Z \in \{40, 55, 70\} \text{ mm}, \quad Y \in \{15, 25, 35\} \text{ mm}, \quad S \in \{0.6, 0.8, 1.0\}.$$

This discretization yields $3 \times 3 \times 3 = 27$ possible actions.

Tripod gait timing was fixed with duty factor 0.5. Feet trajectories in the swing phase were generated using cubic interpolation for smooth lifting and placement. For each action, the commanded foot positions follow the same temporal structure but differ in amplitude via (Z, Y) and timing via S .

Gait Parameter Discretization

Table 1 summarizes the discrete gait parameters used to define the 27-action space.

Parameter	Value 1	Value 2	Value 3
Step height Z (mm)	40	55	70
Stride length Y (mm)	15	25	35
Gait speed S (ratio)	0.6	0.8	1.0

Table 1: Discrete gait parameters defining the 27 available actions.

2.3 State Representation

The state space captures information about the walking surface and is discretized into four terrain classes:

- State 0: hard floor (tile or wood),
- State 1: carpet,
- State 2: cushion or soft mat,
- State 3: unknown or transitional surface.

For each episode, IMU data are sampled at 100 Hz during an initial 0.8 s warm-up phase. Pitch (θ) and roll (ϕ) variance are computed as:

$$\sigma^2 = \text{Var}(|\theta| + |\phi|).$$

Thresholds separating hard, carpet, and cushion surfaces were derived empirically (Table 2).

Surface Type	Variance Range (σ^2)
Hard floor	$\sigma^2 < 0.35$
Carpet	$0.35 \leq \sigma^2 < 0.85$
Cushion	$\sigma^2 \geq 0.85$

Table 2: IMU variance thresholds used to classify terrain states.

Uncertain cases or transitions between surfaces are assigned to State 3.

2.4 Action Index Mapping

For reproducibility, each action index in the Q-table is mapped to a specific (Z, Y, S) triplet. The indexing scheme is summarized in Table 3.

Action Index	Step Height Z (mm)	Stride Y (mm)	Speed S
0–2	40	15	{0.6, 0.8, 1.0}
3–5	40	25	{0.6, 0.8, 1.0}
6–8	40	35	{0.6, 0.8, 1.0}
9–11	55	15	{0.6, 0.8, 1.0}
12–14	55	25	{0.6, 0.8, 1.0}
15–17	55	35	{0.6, 0.8, 1.0}
18–20	70	15	{0.6, 0.8, 1.0}
21–23	70	25	{0.6, 0.8, 1.0}
24–26	70	35	{0.6, 0.8, 1.0}

Table 3: Mapping of action indices to gait parameter combinations (Z, Y, S) .

2.5 Reward Function

After executing a gait for 3–4s, the reward is computed as:

$$R = \Delta x - 0.25 (|\Delta\theta| + |\Delta\phi|), \quad (1)$$

where Δx is the measured forward displacement along the body axis, and $\Delta\theta$, $\Delta\phi$ are RMS deviations in pitch and roll from the initial orientation. Forward displacement is measured using floor markers spaced at 1 cm intervals.

In practice, reward values range from approximately 0 (no progress or highly unstable) to about 3.7 (large displacement with low tilt).

2.6 Q-Learning Framework

The Q-table has shape 4×27 (states $\times$ actions). For each episode transition (s, a, R, s') , the tabular Q-learning update is:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[R + \gamma \max_{a'} Q(s', a') - Q(s, a) \right], \quad (2)$$

with learning rate $\alpha = 0.25$ and discount factor $\gamma = 0.9$.

Action selection follows an ϵ -greedy strategy:

$$\pi(a|s) = \begin{cases} \text{random action,} & \text{with probability } \epsilon_t, \\ \arg \max_a Q(s, a), & \text{with probability } 1 - \epsilon_t, \end{cases}$$

where ϵ_t decays linearly from 0.5 to 0.1 as training progresses.

2.7 Training Procedure

Training data were collected on three surfaces:

- ~ 100 episodes on hard floor,
- ~ 80 episodes on carpet,
- ~ 60 episodes on cushion.

Each episode consists of:

1. Placing the robot on the chosen surface in a neutral pose.
2. Measuring IMU variance for 0.8 s and assigning a terrain state.
3. Selecting an action via the ϵ -greedy policy.
4. Executing the corresponding gait for 3–4 s.
5. Measuring forward displacement and tilt statistics.
6. Computing reward via (1) and updating the Q-table and visitation counts.

The ultrasonic sensor pauses gait execution if an obstacle is detected within 20 cm; episodes in such cases yield smaller Δx and therefore smaller reward.

3 Results

3.1 Learned Q-Value Distribution

The final Q-table exhibits structured clusters of high-valued state–action pairs. Figure 2 visualizes the Q-values for all 27 actions across the four terrain states.

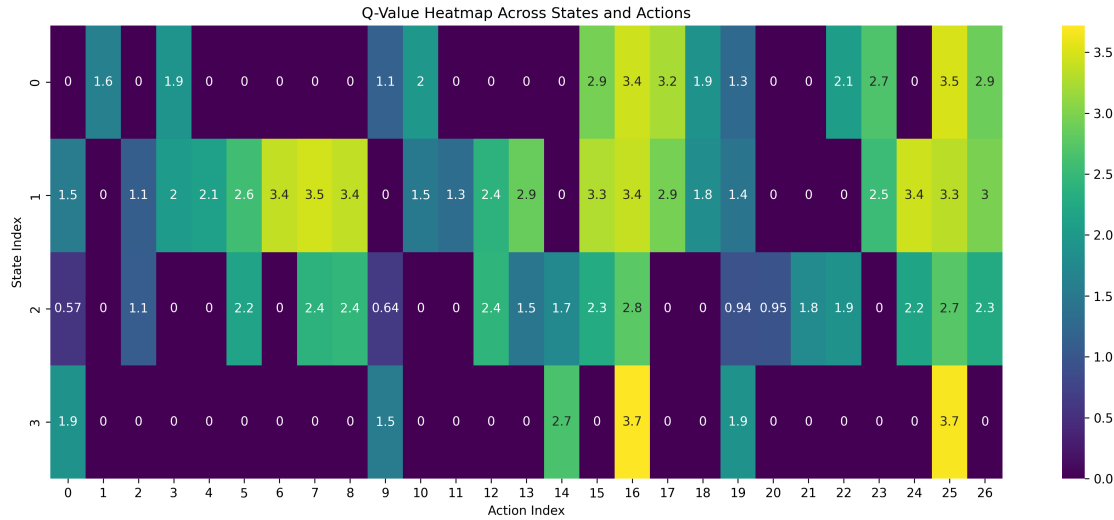


Figure 2: Q-value heatmap across four terrain states (rows) and 27 gait actions (columns). Lighter colors indicate higher expected reward.

The highest Q-values (approximately 3.3–3.7) are concentrated in the region corresponding to actions indexed in the upper right portion of the heatmap. According to Table 3, these actions predominantly correspond to higher step height ($Z = 70$ mm) with medium to large stride length ($Y = 25$ or 35 mm) and moderate gait speed ($S = 0.8$ or 1.0). Low-performing actions ($Q \leq 0.5$) are associated with small step heights and, in some cases, overly aggressive speeds that lead to slipping on soft terrain.

3.2 Forward Displacement Performance

To quantify locomotion performance, a fixed baseline gait (medium Z , medium Y , $S = 0.8$) was compared against the RL-selected actions on each surface. Ten evaluation trials were conducted per surface, and the mean forward displacements are summarized in Table 4.

Surface	Baseline (cm)	RL Policy (cm)	Relative Gain (%)
Hard floor	28.0	37.0	32.1
Carpet	14.0	24.0	71.4
Cushion	8.0	12.5	56.3

Table 4: Measured mean forward displacement for baseline and RL-optimized gaits across three surfaces.

The RL policy provides consistent improvements across all surfaces, with the largest relative gain on carpet. This is consistent with the intuition that moderate compliance introduces slip that can be mitigated by adjusting gait parameters.

3.3 Stability Reduction

Stability was evaluated using the variance of the combined pitch–roll signal during walking. For reporting purposes, baseline tilt variance is normalized to 1.0 on each compliant surface (carpet and

cushion). Table 5 reports the relative reduction achieved by the RL policy.

Surface	Baseline Tilt Var.	RL Tilt Var. (Reduction)
Carpet	1.00	0.88 (12%)
Cushion	1.00	0.77 (23%)

Table 5: Relative reduction in IMU-derived tilt variance when using RL-optimized gaits.

The RL policy reduces oscillation and rocking on soft surfaces, which aligns with the observed preference for high-clearance gaits that avoid excessive penetration into compliant terrain.

3.4 Optimal Actions Identified by RL

To further characterize the learned policy, the top-performing actions were extracted from the Q-table. Table 6 lists representative optimal actions and their associated gait parameters.

Action Index	Z (mm)	Y (mm)	S
21	70	25	0.6
22	70	25	0.8
23	70	25	1.0
24	70	35	0.6
25	70	35	0.8

Table 6: Representative high-performing gait actions identified by RL. These actions exhibit Q-values in the upper range (approximately 3.3–3.7) in compliant terrain states.

These actions share a common structure: large step height and moderate to long stride length. This supports the conclusion that increased foot clearance and stride amplitude are beneficial on compliant surfaces.

3.5 Comparison With Related Work

The observed gains in forward displacement (56–71% on compliant surfaces) are comparable to improvements reported in more complex terrain-adaptive locomotion studies using model predictive control and deep RL (4; 2). Unlike those methods, the present approach:

- operates on a discrete action space with a small Q-table,
- requires no simulation,
- uses only IMU and ultrasonic sensors,
- runs fully on embedded hardware.

This demonstrates that meaningful terrain adaptation can be achieved with modest computational resources, making the approach suitable for educational and low-cost platforms.

4 Conclusions

This study demonstrates that a discrete reinforcement learning controller can substantially improve locomotion robustness of a low-cost hexapod robot across surfaces of varying compliance. The Q-learning agent consistently converged toward high-clearance, moderate-stride gaits that yielded up to 71% improvement in forward displacement and significant reductions in IMU-derived instability on carpet and cushion. The resulting policy is compact, interpretable, and can be deployed without offboard computation.

The methodology was adequate to address the stated problem within the constraints of the platform. Nonetheless, several limitations remain. The state representation relies solely on IMU variance and does not distinguish between frictional properties or slope. The action space is discretized coarsely, potentially excluding more efficient gaits between sampled parameter values. The reward function does not directly incorporate energy consumption or joint stress.

Future work should explore continuous-action RL for finer gait tuning, multimodal terrain classification combining vision and proprioception, and joint torque or current sensing for richer feedback. Additionally, incorporating sim-to-real transfer could enable safer exploration of more aggressive gait strategies prior to deployment on hardware. Overall, the results confirm that simple RL methods can yield meaningful terrain-adaptive locomotion on embedded robotic platforms and provide a practical bridge between classical control and modern deep RL approaches.

References

- [1] D. Dufay and P. Besson, “A survey on hexapod locomotion control,” *Robotics and Autonomous Systems*, vol. 112, pp. 53–67, 2019.
- [2] D. Kim, J. Di Carlo, B. Katz, G. Bledt, and S. Kim, “Highly dynamic quadruped locomotion via whole-body impulse control and model predictive control,” *IEEE Robotics and Automation Letters*, vol. 4, no. 2, pp. 2553–2560, 2019.
- [3] J. Gonzalez and A. Jimenez, “Hexapod gait optimization through parametric analysis,” *International Journal of Advanced Robotic Systems*, vol. 14, 2017.
- [4] J. Hwangbo, J. Lee, A. Dosovitskiy, D. Bellicoso, V. Tsounis, V. Koltun, and M. Hutter, “Learning agile and dynamic motor skills for legged robots,” *Science Robotics*, vol. 4, no. 26, 2019.
- [5] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, “Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor,” *International Conference on Machine Learning*, 2019.
- [6] J. Tan, T. Zhang, E. Coumans, A. Iscen, Y. Bai, D. Hafner, S. Bohez, and V. Vanhoucke, “Sim-to-real: Learning agile locomotion for quadruped robots,” *Robotics: Science and Systems*, 2018.
- [7] J. Kolter and A. Ng, “Learning omnidirectional path following using reinforcement learning,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2008.